LESSON PLAN

HELICOPTER THEORY (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	19
Total No. of Lesson Plans	27	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Forces in climb, Excess power, max angle, maximum rate, Economical climb, time to climb
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Forces in climb, Excess power, max angle, maximum rate, Economical climb, time to climb.
5. Specific Objectives	11.1.3.1 Trainee will be able to describe flight performance of aircraft.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of flight performance charts; Safety: No live flight; Instructor supervision: Monitor chart interpretation accuracy.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the four primary forces acting on an aircraft in steady flight?
(b) How does the relationship between thrust and drag change during a climb?
(c) Define excess power in the context of helicopter performance.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Forces in climb configuration • Excess power definition and calculation • Maximum angle of climb criteria • Maximum rate of climb criteria • Economical climb speed selection	L & Ds	PPT, IB	15 min
(b)	• Time to climb performance factors • Glide range comparison in climb vs descend • Practical application of climb limits • Impact of weight on climb performance • Effect of density altitude on climb	L & Ds	PPT, PDF	20 min
(c)	• Summary of climb performance metrics • Distinction between max angle and max rate • Operational significance of economical climb • Review of time to climb calculations • Preparation for descent forces analysis	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between maximum angle of climb and maximum rate of climb.
(b) How does excess power relate to the ability to maintain a climb?
(c) Why is the economical climb speed different from the maximum rate of climb speed?

12. Home Assignment and Follow Up	Prepare a brief note comparing the forces in climb with the forces in a descent glide, referencing the next lesson plan (LP 20).

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Descend forces in a descend glide range.	Refer to the master syllabus reference list	LP 20

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min